

WEEK 2 – DATA SCIENCE AND VISUALIZATION

Dataset Representation and Initial Exploration

Project: Accident Data Analysis

Dataset: AccidentsBig.csv

1. Introduction

Week 2 focuses on the initial representation and exploration of the raw accident dataset. This stage establishes an understanding of the dataset structure before performing data cleaning and deeper exploratory analysis. The dataset is examined using Python and Pandas to understand its dimensions, attributes, data types, missing-value presence, duplicates, and basic statistical characteristics.

2. Objectives

The main objectives of Week 2 were to load and understand the accident dataset, inspect its dimensions and structure, identify numerical and categorical attributes, represent missing values and duplicate records, examine descriptive statistics, study unique values, and create a basic representation of accident severity.

3. Dataset Description

The project uses **AccidentsBig.csv**, a structured accident dataset containing multiple records and attributes related to accidents. The attributes include numerical and categorical information that can be used for understanding accident characteristics and for subsequent data-cleaning and exploratory-analysis stages.

4. Tools and Technologies Used

Python, Pandas, NumPy, Matplotlib, Seaborn, and Jupyter Notebook were used for the initial dataset representation and exploration.

5. Dataset Loading

The raw dataset was loaded into a Pandas DataFrame. Each row represents an accident-related record, while each column represents an attribute describing that record.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("AccidentsBig.csv", low_memory=False)
```

6. Dataset Dimensions

The number of rows and columns was examined using the DataFrame shape. This provides an immediate understanding of the size of the raw dataset.

```
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])
print("Dataset shape:", df.shape)
```

7. Dataset Preview

The first and last records were displayed to understand how the accident records are organized and to inspect the values stored in the different attributes.

```
df.head()

df.tail()
```

8. Column Names and Data Types

The column names and data types were examined to distinguish numerical variables from categorical or text-based attributes. The dataset contains a mixture of numerical and categorical information.

```
for i, col in enumerate(df.columns, start=1):
    print(i, col)

df.info()
```

9. Numerical and Categorical Attributes

The dataset was separated conceptually into numerical and categorical/text attributes. Numerical columns can be summarized statistically, while categorical attributes can be examined through frequencies and unique-value counts.

```
numerical_columns = df.select_dtypes(include=np.number).columns.tolist()
categorical_columns = df.select_dtypes(include=["object", "category"]).columns.tolist()

print("Numerical columns:")
print(numerical_columns)

print("\nCategorical/Text columns:")
print(categorical_columns)
```

10. Missing-Value Representation

Missing values were identified and quantified as part of the initial data-quality assessment. At this stage, the values were only represented; no imputation or deletion was performed. Detailed missing-value treatment is reserved for the data-cleaning stage.

```
missing_values = df.isnull().sum()
missing_percentage = (missing_values / len(df)) * 100

missing_summary = pd.DataFrame({
    "Missing Values": missing_values,
    "Missing Percentage": missing_percentage.round(2)
}).sort_values("Missing Values", ascending=False)

missing_summary
```

11. Duplicate Record Check

Duplicate records were checked to establish a baseline for the raw dataset. The result can later be compared with the cleaned dataset in Week 3.

```
duplicate_count = df.duplicated().sum()
print("Number of duplicate records:", duplicate_count)
```

12. Descriptive Statistics

Descriptive statistics were generated for the numerical attributes. The summary includes count, mean, standard deviation, minimum, quartiles, and maximum values, providing an initial understanding of the numerical variables.

```
df.describe().T
```

13. Unique-Value Representation

The number of unique values in each column was examined to understand the variety of values present in the dataset and to identify attributes that may represent categorical information.

```
unique_summary = pd.DataFrame({
    "Column": df.columns,
    "Unique Values": [df[col].nunique(dropna=True) for col in df.columns]
})

unique_summary
```

14. Accident Severity Representation

The accident-severity attribute, when available in the dataset, was represented using frequency counts and a basic count plot. This provides an initial view of how records are distributed across severity categories.

```
if "Accident_Severity" in df.columns:
    print(df["Accident_Severity"].value_counts(dropna=False))

plt.figure(figsize=(8, 5))
sns.countplot(data=df, x="Accident_Severity")
plt.title("Accident Severity Distribution")
plt.xlabel("Accident Severity")
plt.ylabel("Number of Records")
plt.show()
```

15. Initial Observations

The Week 2 representation establishes the basic structure and data quality characteristics of the accident dataset. The dataset contains multiple accident-related attributes with a combination of numerical and categorical information. Missing values and duplicate records were identified for further treatment. Descriptive statistics provide a baseline for numerical variables, while unique-value and severity representations provide an initial understanding of important categorical information.

16. Preparation for Further Analysis

The Week 2 analysis provides the foundation for the next stage of the project. The raw dataset can now proceed to data cleaning, where missing values, inconsistent data, duplicates, and potential outliers will be examined and handled appropriately.

17. Conclusion

Week 2 successfully established a structured representation of the raw accident dataset. The dataset was loaded and examined through its dimensions, records, columns, data types, numerical and categorical attributes, missing-value counts, duplicate records, descriptive statistics, unique values, and accident-severity distribution. This initial exploration provides a clear baseline for the Week 3 cleaning and data-quality assessment stage.

18. Project Status

Week 2: Dataset Representation and Initial Exploration — Completed